

Synthetic Guide / EPG Enterprise Backend

Regression Fixture v1

Purpose. This document is intentionally written as an enterprise architecture reference using fictional internal product names. It tests whether an architecture-understanding engine can infer component meaning from explicit definitions, relationships, numbered journeys and deployment evidence rather than relying only on familiar technology brands.

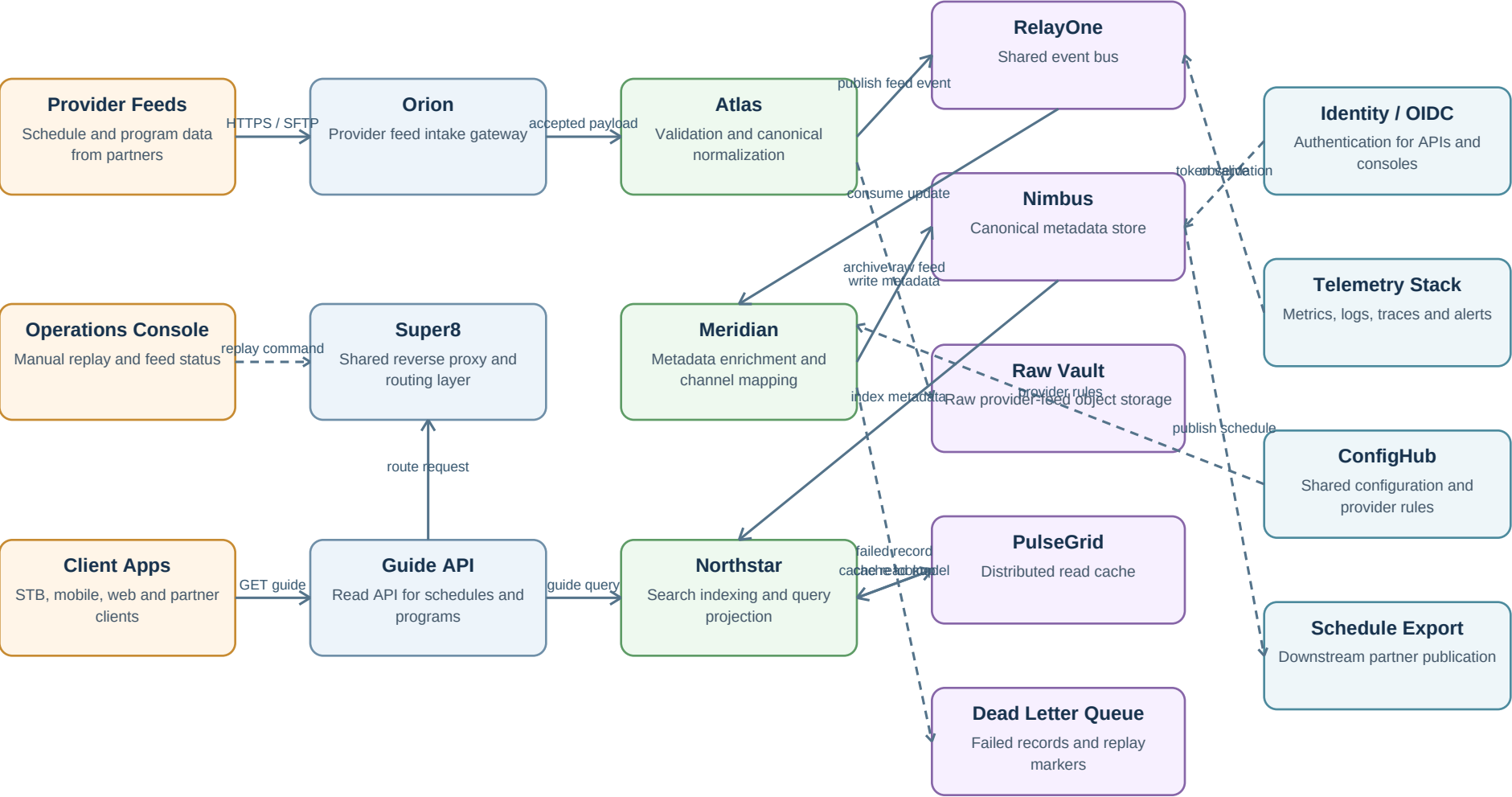
System scope. The platform ingests electronic program guide (EPG) and schedule data from external providers, validates and normalizes records, enriches metadata, publishes update events, serves guide queries to client applications, handles failed records, and exports schedules to downstream partners.

Regression objective	Expected evidence
Document-defined internal names	Super8, Orion, Atlas, Meridian, RelayOne, Nimbus, Northstar, PulseGrid, ConfigHub and Raw Vault are defined in prose.
Multiple journey types	Provider ingestion, guide query, event propagation, failure/replay and export.
Enterprise deployment	Three availability zones with repeated runtime families and a separate shared platform layer.
Shared infrastructure	RelayOne, Nimbus, Raw Vault, Identity / OIDC, Telemetry Stack, ConfigHub and Schedule Export serve multiple zones.
Safety expectations	Do not infer active-active, failover, ownership or disaster recovery unless explicitly stated.

Synthetic only. No real company names, proprietary schemas, customer data or production topology are included.

Guide / EPG Enterprise Backend - Canonical Architecture

Internal product names are synthetic; responsibilities are explicit in the document.



Numbered journeys represented in this architecture:

- 1) Provider feed ingestion 2) Guide query 3) Update propagation 4) Failure and replay 5) Partner schedule export

Component Definitions

Orion	Provider feed intake gateway. Accepts schedule and program feeds over HTTPS or SFTP, authenticates the source, records receipt metadata and forwards accepted payloads to Atlas.
Super8	Shared reverse-proxy and request-routing layer. Routes operations-console requests and client guide requests to the appropriate internal API. It is not a media packager or database.
Atlas	Validation and canonical normalization service. Checks required fields, timestamps, channel identifiers and program references, then converts provider-specific payloads into the canonical guide format.
Meridian	Metadata enrichment and channel-mapping service. Joins normalized schedules with channel, station, artwork, genre and regional mapping data before writing Nimbus.
RelayOne	Shared event bus. Distributes guide-update events to Northstar, PulseGrid, Schedule Export and operational subscribers.
Nimbus	Canonical metadata store. Persists channels, programs, airings and schedule relationships as the durable system of record.
Raw Vault	Object storage for original provider payloads and replayable feed archives.
Northstar	Search indexing and query-projection service used by Guide API.
PulseGrid	Distributed read cache for frequently requested guide windows and program summaries.
ConfigHub	Shared configuration service containing provider rules, schema versions, channel mappings and feature flags.
Dead Letter Queue	Shared holding area for records that fail validation or enrichment, including a reason code and replay marker.
Schedule Export	Shared outbound publication service that transforms canonical schedules into partner-specific formats.
Telemetry Stack	Shared observability platform for metrics, logs, traces and alerting across all services and zones.
Identity / OIDC	Shared authentication and token-validation service for operations consoles, partner APIs and administrative requests.

Numbered System Journeys

Journey 1 - Provider feed ingestion

1. Provider Feeds send schedule data to Orion. 2. Orion authenticates the provider and records receipt metadata. 3. Orion sends the accepted payload to Atlas. 4. Atlas validates required fields and normalizes the provider format. 5. Atlas archives the original payload in Raw Vault and publishes a normalized-feed event to RelayOne. 6. Meridian consumes the event, enriches the records, and writes canonical metadata to Nimbus. 7. Meridian publishes a guide-update event to RelayOne.

Journey 2 - Client guide query

1. A set-top box, mobile app or web client sends a guide query to Super8. 2. Super8 routes the request to a healthy Guide API instance in one availability zone. 3. Guide API validates the token through Identity / OIDC when required. 4. Guide API checks PulseGrid. 5. On a cache miss, Guide API reads canonical records from Nimbus and query projections from Northstar. 6. Guide API returns channels, programs and airings.

Journey 3 - Update propagation

1. Meridian commits enriched metadata to Nimbus. 2. Meridian publishes a guide-update event to RelayOne. 3. Northstar consumes the event and updates search projections. 4. PulseGrid consumers invalidate or refresh affected cache keys. 5. Schedule Export prepares approved partner outputs. 6. Telemetry Stack records processing latency, failures and event lag.

Journey 4 - Failure and replay

1. Atlas or Meridian detects a validation or enrichment failure. 2. The failing record is sent to the Dead Letter Queue with a reason code and source-feed reference. 3. Operations Console displays the failure through Super8. 4. An operator corrects a provider rule in ConfigHub or approves replay. 5. Raw Vault returns the payload to Atlas. 6. Successful replay follows the normal ingestion path.

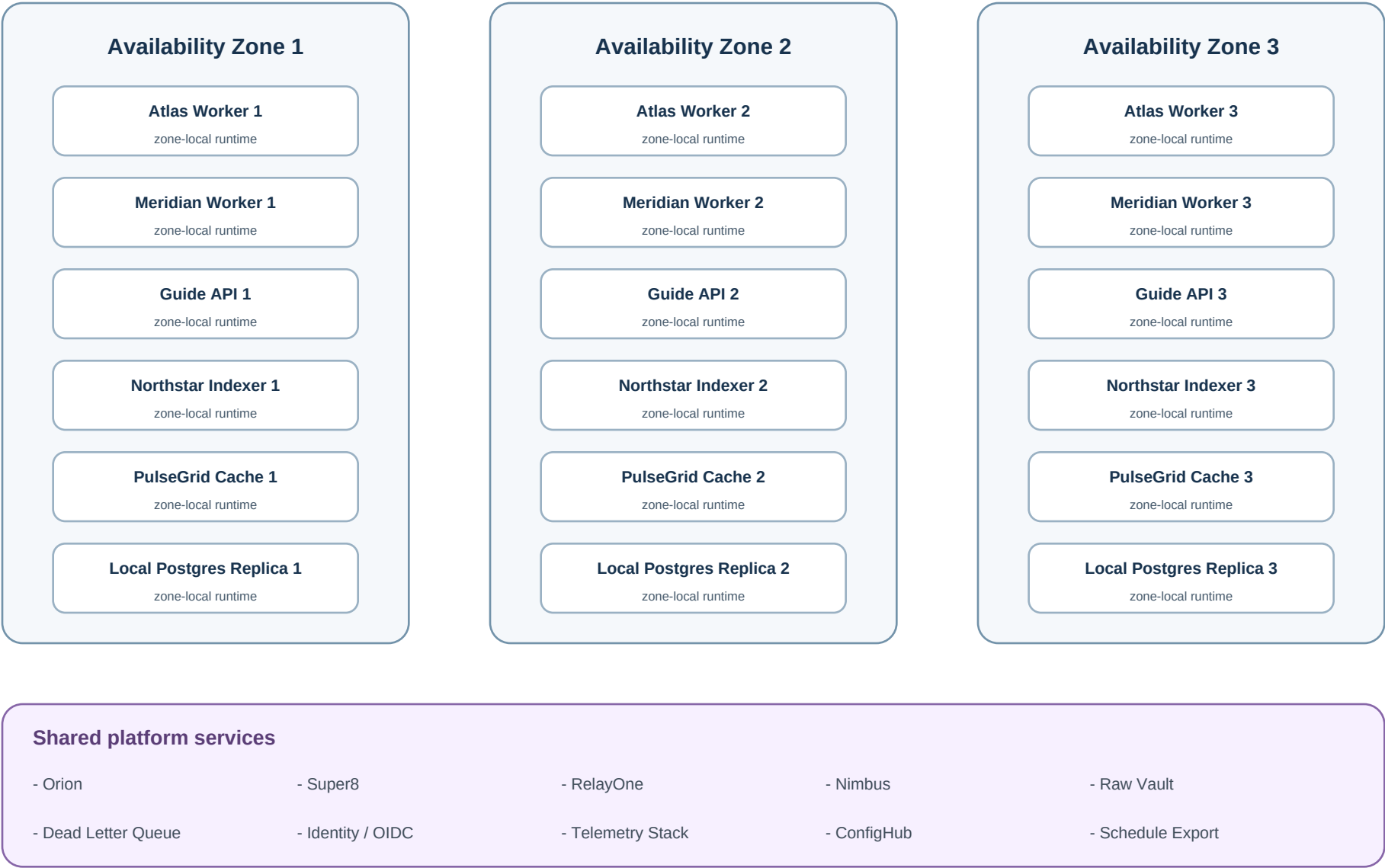
Journey 5 - Partner schedule export

1. RelayOne publishes a guide-update event after Nimbus is updated. 2. Schedule Export consumes the event and reads the canonical schedule from Nimbus. 3. It applies partner-specific rules from ConfigHub. 4. The formatted schedule is sent to the approved downstream partner. 5. Telemetry Stack records delivery status and alerts on repeated failures.

Journey hint: ingestion and propagation are event-driven; guide query is synchronous; failure/replay is validation/control; export is outbound integration.

Deployment Topology - Three Availability Zones

Zones contain replicated workloads. Shared platform services sit outside the zones and serve all three.



Deployment and Evidence Notes

Explicit deployment units	Availability Zone 1, Availability Zone 2 and Availability Zone 3 are explicit deployment boundaries.
Replicated runtime families	Atlas Worker 1/2/3, Meridian Worker 1/2/3, Guide API 1/2/3, Northstar Indexer 1/2/3, PulseGrid Cache 1/2/3 and Local Postgres Replica 1/2/3 are deployment-local instances. The generic family name is not one global singleton.
Shared platform services	Orion, Super8, RelayOne, Nimbus, Raw Vault, Dead Letter Queue, Identity / OIDC, Telemetry Stack, ConfigHub and Schedule Export sit outside the three zones and serve workloads in more than one zone.
Topology safety	The diagram shows multi-zone deployment and shared services. It does not claim active-active operation, failover behavior, disaster recovery, traffic percentages or ownership.
Relationship evidence	Solid arrows represent runtime data or request flow. Dashed arrows represent control, configuration, authentication or operational support interactions.
Expected engine behavior	Preserve compound names, reject glossary/list summaries as components, distinguish local replica families from shared infrastructure, identify multiple journeys, and retain evidence-backed explanations.

Expected shared-infrastructure candidates: RelayOne, Nimbus, Raw Vault, Dead Letter Queue, Identity / OIDC, Telemetry Stack, ConfigHub and Schedule Export. Orion and Super8 may also qualify if graph evidence shows they serve multiple deployment units. Zone-local replicas must not be promoted as shared infrastructure.